

STATISTICAL SHAPE MODELLING

What is Statistical Shape Modelling?

- A **statistical shape model (SSM)** captures how the shape of a bone varies across a population.
- The model is built from hundreds or thousands of 3D bone scans and learns the most common patterns of anatomical variation.
- Statistical methods can then be used to predict the shape of an individual's bones from a limited set of measurements.

What can we learn from bone shape?

- Bone shape varies between individuals due to factors such as **body size, age, sex, genetics, and activity history**.
- Statistical shape models help researchers understand how anatomy differs across people and populations.
- They can reveal subtle shape variations that are difficult to measure using traditional anatomical measurements.
- These models are widely used in **biomechanics, orthopaedics, and medical device design**.

Why is this useful?

- Personalised bone models help researchers study how people move.
- They can be used in musculoskeletal simulations to estimate muscle and joint mechanics.
- Rapid anatomy prediction can reduce the need for expensive medical imaging.
- These techniques are helping make personalised healthcare and biomechanical modelling more accessible.

